# Tag, You're Built!

**Nothing worked the first time. Or the second.**

Here's how we built the infrastructure for our laser tag event.

We started by planning the event. Our team sat down and asked the important questions to create a blueprint of everything that was needed. Once we had a clear idea of the elements required to make the game possible, we settled on the themes. Planning the flow early on turned out to be our smartest decision because it helped make a rather complicated ordeal simpler and more organised.

The heart of the event was the blasters. We figured that if we got the blasters right, the hardest part would be sorted. Since we were building everything inhouse, that meant the blasters too, so we started with the design, covering both the exterior and the electronics. We drew up the blueprint for the prototype on Fusion 360 and 3D printed it in our design labs. The electronic design had to be done alongside it because the boards and wires needed to fit properly inside the chassis. We designed it around the ESP32, which was the brain of the system. We also wrote the software alongside it, covering game control, connections, and scoring logic. It took a lot of trial and error, but slowly, the hardware and software started coming together on paper.

Our budget was slim, and building around the ESP32 was no small ordeal. All the sensors and boards had to be accounted for, and even a single wrong order could have been catastrophic for the event. We couldn't afford any mistakes, so we stayed frugal at every step, planning everything out before executing it. Building everything inhouse helped, as did planning and choosing our parts wisely.

## And then the shipment got delayed.

Everything we had planned depended on that one delivery of ESP32s. Without them, nothing would work because they were the brains of the operation. When the delivery slipped, so did our schedule. There were multiple delays and a lot of sleepless nights spent trying to figure out what to do next.

When the parts finally arrived, the work began. We assembled the electronics and circuitry, which was pretty complex, then ran the software and tested everything. This took a lot of trial and error, but we finally managed to build a working prototype.

We spent time building redundancies to account for as many potential failures as possible. We also wanted the blasters to be comfortable to use so that the participants could have a great time with them.

After we had the core infrastructure up and running, the only things left were the decor and some last minute preparations, and it was finally ready.

Creating this event took a lot of time and effort from both clubs. It was tiring and stressful, but also a lot of fun and rewarding. Now, we hope everyone who plays enjoys it as much as we enjoyed building it.